Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination 2024
Course Code: EEE 455, Session: 2020-21
Course Title: VLSI Design and Testing

Full Marks: 100 **Credits: 3.0** **Duration: 3 Hours**

Group-A

[Answer any two questions from Group-A]

1. (a) Explain the C-V characteristics of intrinsic MOS gate capacitance in different regions. 4

   (b) A pseudo NMOS inverter is designed such that the output voltage becomes 0.25 V when the input voltage is 5 V. The following data are given: $\mu_n C_{ox} = 120\ \mu$A/V^2, $\mu_p C_{ox} = 50\ \mu$A/V^2, $V_{ton} = 1$V, $V_{top} = -1$ V, $V_{DD} = 5$ V, $\gamma = 0.5$. Assume that the body of the NMOS transistor is connected with the ground and the body of the PMOS transistor is connected with V_{DD}.

   (i) Draw the circuit diagram of the inverter.
   (ii) Calculate the inverter ratio i.e., the ratio of driver transistor (W/L) to the load transistor (W/L).
   (iii) What will be the output voltage if the input voltage is held at 0.25 V? Explain your answer. 8

   (c) Draw the schematic and stick diagram of the circuit implementing the logic $Y = \overline{(A + B + C) \cdot D}$ 6

   (d) Estimate the minimum delay of the path from A to B in Figure 1(d) and choose transistor sizes to achieve this delay. The initial NAND2 gate may present a load of 8 λ of transistor width on the input and the output load is equivalent to 45 λ of transistor width. 5

   (e) Differentiate between the diffusion process and the ion implantation process in semiconductor fabrication. 2

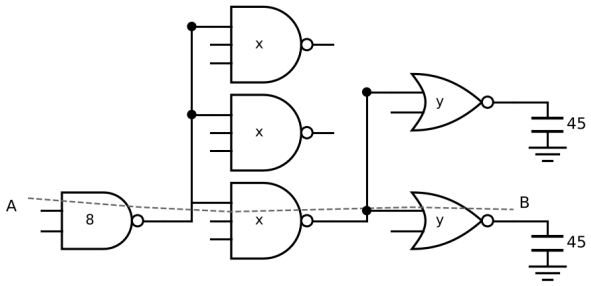


Figure 1(d) of the paper (redrawn)

2. (a) For the RC circuit shown in Figure 2(a), compute the Elmore delays from input node to node 7 and from input node to node 5 of this circuit in terms of R and C component values. 5

   (b) Briefly explain what happens to the transfer characteristics and noise margins when beta ratio is increased in a CMOS inverter pair. 6

   (c) Design an 8-input AND gate using FO4 in different configurations. Determine which configuration provides the fastest performance. 7

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(d) Write short note on –i) Static power dissipation ii) Dynamic power dissipation. 4

(e) What is domino gate? Draw the circuit diagram of a domino OR gate. 3

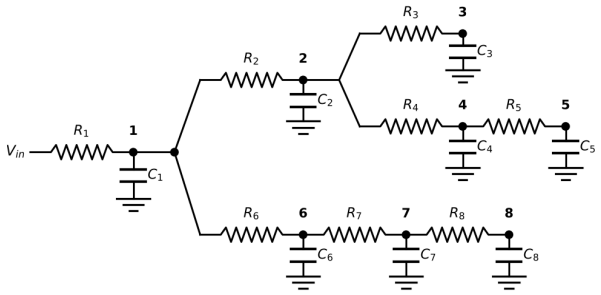


Figure 2(a) of the paper (redrawn)

3. (a) You have to design a car alarm system in which the alarm will ring (Y= 1) if anyone of the following condition is satisfied.

(i) The key is in (K= 1), The Seat belt is fastened (B=1) and the door is open (D=0) or
(ii) The key is in, Door is closed and Seat belt is not fastened.

Write the Truth table, the Boolean equations and show the implementation of the circuit in (i) static CMOS, (ii) NMOS, (iii) Pseudo NMOS, (iv) Footed dynamic CMOS and (v) Un-footed dynamic CMOS. 6

(b) For the logic function $Y = \overline{(A + BC) \cdot D}$, choose transistor widths to achieve equal rise and fall resistance as a unit inverter. Assume, $\mu_n = 3\mu_p$ 4

(c) Show that the dynamic power dissipation of a CMOS inverter varies directly with square of the power supply voltage whereas the delay of the circuit varies inversely with the power supply voltage. 8

(d) Explain the inverter width sizing for favoring the rising edge, falling edge, or achieving an unskewed inverter. 7

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Group-B

[Answer any two questions from Group B]

4. (a) In the following circuit of Figure 4(a), determine whether there is any set up or hold time violation. 10

   (b) Show the circuit diagram of an asynchronous resettable Flip-Flop and explain its operation utilizing a timing diagram. 6

   (c) Suppose you have written Verilog code to operate a microprocessor. What will be the upcoming steps till its fabrication? Briefly explain your answer. 6

   (d) Implement a 2-input AND gate using both transmission gates and pass-transistor logic. 3

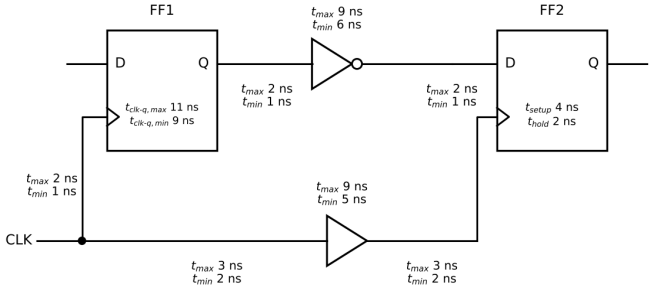


Figure 4(a) of the paper (redrawn)

5. (a) (i) Show the circuit diagram of a 2 x 2 SRAM array which uses 6-transistor memory cell as basic storage cell. Clearly show the row select, column select, pre-charge and sense signal in your circuit.

   (ii) Explain how the READ and WRITE operation are performed showing the second row and first column cell as an example.

   (iii) Explain how sense amplifier reduces the READ time of the cell. 10

   (b) A 4-bit datapath consists of register, ALU and a shifter. Show two possible bus architectures of the system such that an addition operation of two operands stored in the register and storing the result back in the register can be computed in at most two clock cycles. 7

   (c) Explain the following: (i) positive mask, (ii) negative mask, (iii) positive photoresist and (iv) negative photoresist. 2

   (d) Describe the process sequence for patterning SiO_2 as illustrated in the figure 5(d). 6

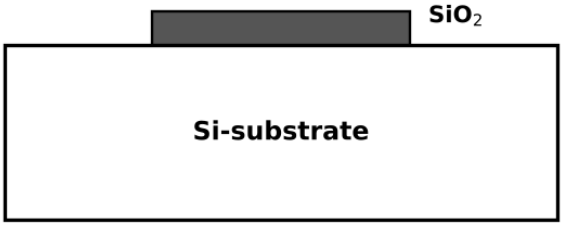


Figure 5(d) of the paper (redrawn)

6. (a) Design a 2-to-1 multiplexer using pass transistor logic. Mention its working procedure. 4

   (b) Design a 7-segment display driver in Verilog. 5

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(c) You have to design a divide-by-3 counter using FSM.

i) Draw the state transition diagram of the finite state machine described.
ii) Derive the state transition table and equations for the next-state logic.
iii) Draw the complete FSM circuit. 6

(d) Write a Verilog code for the FSM described in Question 6(c). 3

(e) Sketch a dot diagram for a 2-input XOR using a PLA. 3

(f) Fill in the truth table for the flip flop shown in Figure 6(f). Is this a positive edge triggered flip flop? 4

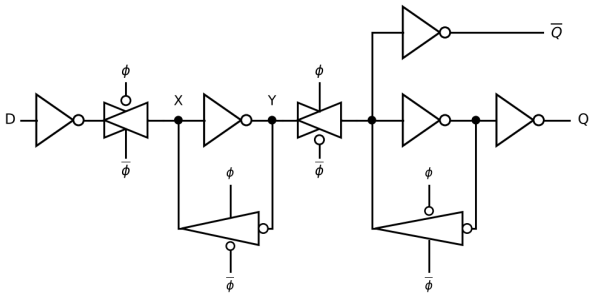


Figure 6(f) of the paper (redrawn)

ϕ	$\overline{\phi}$	D	X	Y	$\overline{Q}$	Q
0	1		1			1
1	0	1				
0	1	1				
1	0	1				

The table of Figure 6(f), as printed

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